Certificate-Only Causal Discovery: Monitoring-Valid Edges and Predictive-Fit Orientation

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Abstract

Constraint-based causal discovery deletes an edge when a conditional-independence test fails to reject, so a lack of power is reported as a conclusion, and its output cannot be monitored as data arrive: an analyst who reruns it at each sample size and keeps any edge ever reported accumulates false edges. We study certificate-only discovery, which asserts a feature of the graph only by rejecting a null that is true whenever the feature is absent. The adjacency certificate is the minimum over conditioning sets of sequential tests that a partial regression coefficient is zero. It bounds the probability that a false edge is ever certified, uniformly over time and without faithfulness; the bound is exact under Gaussian errors and asymptotic otherwise. The orientation certificate compares sequentially fitted forecasters for the two factorisations of a pair and certifies a direction only when two model classes agree, so what it certifies is better predictive fit, which identifies the causal direction in additive-noise models. In 150 random graphs monitored from $n = 100$ to **6400**, CERT-CD reported a false edge in **2.7%** of graphs under Laplace noise, against **27%** to **91%** for PC, GES and DirectLiNGAM, at the price of leaving most non-adjacent pairs undecided. On the Tübingen cause-effect pairs, under a protocol fixed before the run, the orientation certificate committed on **12** of **80** scorable pairs and was right on **11**. Both certificates fail outside their models, and we report where.

Keywords: causal discovery, anytime-valid inference, e-values, sequential testing, additive noise models, family-wise error control

MSC Classification: 62L12 , 62H22 , 62F03

1 Introduction

A causal discovery algorithm returns a graph, and a practitioner reading it wants to know which of its features the data support. Constraint-based methods such as PC and FCI (Colombo, Maathuis, Kalisch, & Richardson, 2012; Spirtes, Glymour, & Scheines, 2000; Spirtes, Meek, & Richardson, 1995) start from the complete graph and delete an edge whenever a conditional-independence test fails to reject. Failure to reject is compatible with independence, with a dependence too weak to see at the current sample size, and with a test that had little power; the deletion happens regardless. Two consequences follow. Finite-sample correctness requires faithfulness or a quantitative strengthening of it (Robins, Scheines, Spirtes, & Wasserman, 2003; Uhler, Raskutti, Bühlmann, & Yu, 2013), because only such an assumption turns "the test did not reject" into "the variables are independent". And the output is two-valued where the evidence is three-valued: a missing edge means either "the data indicate no edge" or "nothing was established about this pair".

There is a third consequence that matters in practice. Data accumulate, and an analyst who inspects the estimate as they do, and keeps any edge that has ever appeared, is performing optional stopping. A fixed-sample guarantee does not cover this. Section 4.2 measures the effect on eight-node graphs: when PC, GES and DirectLiNGAM are rerun at each sample size from 100 to 6400, between a quarter and nine tenths of the graphs have at least one false edge in the union of the estimates.

This paper studies the opposite discipline, which we call *certificate-only* discovery: assert a feature of the graph only by rejecting a null hypothesis that is true whenever the feature is absent, and never assert anything by failing to reject. An algorithm that obeys the rule starts from the empty graph and adds; it cannot report an edge as absent, only as undecided; and what it controls is the probability of asserting something false. If each test is an anytime-valid process, the guarantee holds at every sample size at once and survives continuous monitoring.

Contributions.

1. *Adjacency certificates under monitoring.* For a linear structural equation model with independent errors, d-separation implies that a partial regression coefficient is zero, so adjacency has a certifying null that needs no faithfulness and no Gaussian errors. The minimum over conditioning sets of per-set sequential tests bounds the probability that a false edge is *ever* certified (Section 3.3). The bound is exact for Gaussian errors and asymptotic otherwise, and we measure how far from nominal it is off-model (Section 4.3).
2. *An orientation certificate based on predictive fit.* For a pair of variables we fit forecasters for the factorisations $x \to y$ and $y \to x$ sequentially, and certify a direction when the running mean of the log-score difference has a confidence sequence that excludes zero under two model classes at once (Section 3.4). The claim is about predictive fit and is valid whether or not either model is correct; it identifies the causal direction under additive-noise models.
3. *Evidence under pre-specified protocols.* We compare against PC, GES and DirectLiNGAM under monitoring (Section 4.2), evaluate the orientation certificate

on simulated pairs and on the Tübingen cause-effect pairs with a protocol committed before the run (Sections 4.5 and 4.6), and report the regimes where the certificates fail (Section 4.7).

Scope.

The method targets continuous, observational data from a causally sufficient system whose mechanisms are linear or additive-noise, at sample sizes of a few hundred or more. We chose the benchmarks to match this domain and fixed the selection criteria before running; data outside it are reported as a limitation, not dropped.

Outline.

Section 2 places the work in the literature. Section 3 develops the certificates and the algorithm. Section 4 reports the experiments and Section 5 discusses limitations. Proofs and further results are in the appendices.

2 Related work

Error control in constraint-based discovery.

PC-p (Strobl, Spirtes, & Visweswaran, 2019) bounds an edge's p-value by the maximum p-value over conditioning sets and controls the false discovery rate of edge and orientation decisions with Benjamini–Yekutieli; it is finite-sample but valid at a fixed sample size, and its orientation rules operate within the Markov equivalence class. The conservative PC algorithm of Ramsey, Spirtes, and Zhang (2006) marks ambiguous triples instead of resolving them. Faithfulness-based guarantees break down when parameters are near a cancellation (Robins et al., 2003; Uhler et al., 2013; Zhang & Spirtes, 2008). ϵ-CUT (Shaska & Mitra, 2025) gives one-sided false-edge control without faithfulness at each fixed sample size; Csillag, Struchiner, and Goedert (2025) calibrate batched Fisher-z statistics into per-test e-values inside PC. The certificate-only discipline here is the same one-sided logic made time-uniform and extended to orientation.

Anytime-valid inference.

Ville's inequality (Ville, 1939) underlies confidence sequences (Darling & Robbins, 1967; Howard, Ramdas, McAuliffe, & Sekhon, 2021; Robbins, 1970) and the e-value literature (Grünwald, de Heide, & Koolen, 2024; Ramdas, Grünwald, Vovk, & Shafer, 2023; Vovk & Wang, 2021). Asymptotic confidence sequences (Waudby-Smith, Arbour, Sinha, Kennedy, & Ramdas, 2024) and anytime-valid linear models (Lindon, Ham, Tingley, & Bojinov, 2026) extend the approach to regression coefficients under weak moment conditions. Conditional-independence testing is intrinsically hard: no test with uniform level over all conditional-independence nulls has power against any alternative (Shah & Peters, 2020). Sequential tests therefore either assume a known conditional law (Grünwald, Henzi, & Lardy, 2024; Shaer, Maman, & Romano, 2023) or restrict the null, as we do by testing a partial regression coefficient.

Orientation.

Linear non-Gaussian models identify the full graph (Shimizu, Hoyer, Hyvärinen, & Kerminen, 2006) and DirectLiNGAM (Shimizu et al., 2011) and the pairwise likelihood ratios of Hyvärinen and Smith (2013) estimate it; Shimizu and Kano (2008) test direction using non-normality. Additive-noise models (Hoyer, Janzing, Mooij, Peters, & Schölkopf, 2009; Peters & Bühlmann, 2014) identify direction for nonlinear mechanisms, and location-scale models extend this to heteroscedastic noise (Immer et al., 2023; Xu, Mian, Marx, & Vreeken, 2022). Compression- and quantile-based scores (Marx & Vreeken, 2017; Tagasovska, Vatter, & Chavez-Demoulin, 2020) perform well on the Tübingen pairs (Mooij, Peters, Janzing, Zscheischler, & Schölkopf, 2016). Linear methods fit those pairs poorly (Schkoda & Drton, 2025), and a test that is valid against a misspecified model does not control direction error. Methods that abstain exist: Prakash, Xia, and Erosheva (2024) and Ding and Zhang (2025) return an inconclusive outcome, Prakash, Xia, and Erosheva (2026) gate on a linearity check, and Uehara (2026) attach per-edge resolved or impossible codes in an interactive protocol. None of these gives a time-uniform error statement; ours is a statement about predictive fit that holds without either model being correct (Section 3.4), a comparison in the spirit of Vuong (1989) and of sequential forecaster comparison (Choe & Ramdas, 2024).

3 Method

3.1 Setting and assumptions

Let $\mathcal{G}^\star$ be a directed acyclic graph on $V = \{1, \dots, d\}$ and let $X = (X_1, \dots, X_d)$ follow a linear structural equation model $X = B^\top X + e$, where B is supported on the edges of $\mathcal{G}^\star$ and the errors $e_1, \dots, e_d$ are mutually independent with finite fourth moments. Observations $o_1, o_2, \dots$ are i.i.d. draws of X and arrive in batches; $\mathcal{F}_t$ is the σ-field of the first t observations. We write $\mathrm{pa}(i)$ and $\mathrm{de}(i)$ for the parents and descendants of i and $\mathcal{S}_k(i,j) = \{S \subseteq V \setminus \{i,j\} : |S| \le k\}$ for the admissible conditioning sets, a family fixed before any data are seen.

Assumption 1 (Separator size) Every non-adjacent pair of $\mathcal{G}^\star$ has a d-separating set of size at most k.

Assumption 1 is structural. It holds whenever k is at least the maximum in-degree of $\mathcal{G}^\star$, because for non-adjacent i, j either $\mathrm{pa}(i)$ or $\mathrm{pa}(j)$ d-separates them; $k = d - 2$ makes it vacuous at an exponential computational cost.

For a pair (i, j) and a set S let $\beta_{i \leftarrow j \cdot S}$ denote the coefficient of X_j in the population regression of X_i on $(1, X_j, X_S)$.

Lemma 1 *If S d-separates i and j in $\mathcal{G}^\star$, then $\beta_{i \leftarrow j \cdot S} = 0$.*

This is the standard fact that d-separation implies a vanishing partial covariance in a linear structural equation model (Spirtes et al., 2000). It is a moment condition, weaker than conditional independence, and it does not require Gaussian errors. Faithfulness is the converse and is not assumed.

3.2 Certificates

Definition 1 Let ϕ be a feature of $\mathcal{G}^\star$. A *certifying null* for ϕ is a hypothesis H_ϕ that holds whenever ϕ is false. A process $(E_t)_{t\geq 0}$ is *Ville-valid* for H if $E_t \geq 0$ and $\mathbb{P}(\exists t : E_t \geq \lambda) \leq 1/\lambda$ for every $\lambda \geq 1$ and every $P \in H$. A *certificate for ϕ at level α* is the event that a Ville-valid process for H_ϕ reaches $1/\alpha$.

Proposition 2 (Soundness) *If ϕ is false, the probability that it is ever certified at level α is at most α.*

The proof is the definition: if ϕ is false then $P \in H_\phi$. The implication runs one way. H_ϕ may also hold when ϕ is true, and the certificate then never fires; this is what buys validity without faithfulness and what costs power. Three consequences follow. There is no certifying null for "i and j are non-adjacent", so the algorithm cannot delete and must start from the empty graph. Its output is three-valued (certified adjacent, certified with a direction, undecided). And power is descriptive: nothing guarantees that a true edge is ever certified.

Ville-valid is weaker than *e-process*, which requires $\mathbb{E}[E_\tau] \leq 1$ at every stopping time of a single filtration. All our bounds use only the event that a process ever reaches a level, so they hold at every data-dependent stopping time of the run, but we do not claim the stronger property. Section 3.3.1 shows why.

3.3 The adjacency certificate

For a pair (i, j) consider

$$H_{ij}^{\text{sep}} : \quad \exists S \in \mathcal{S}_k(i,j) \;\text{ with }\; \beta_{i \leftarrow j \cdot S} = 0. \tag{1}$$

By Lemma 1 and Assumption 1, H_{ij}^{sep} holds for every non-adjacent pair, so it is a certifying null for adjacency. Let $E_t^{(S)}$ be a Ville-valid process for the point null $\beta_{i \leftarrow j \cdot S} = 0$ and define

$$A_t(i,j) = \min_{S \in \mathcal{S}_k(i,j)} E_t^{(S)}(i,j). \tag{2}$$

Proposition 3 *Let $\lambda \geq 1$ and let (i, j) be non-adjacent. Then $\mathbb{P}(\exists t : A_t(i,j) \geq \lambda) \leq 1/\lambda$.*

Proof Choose $S^\star \in \mathcal{S}_k(i,j)$ with $\beta_{i \leftarrow j \cdot S^\star} = 0$. Since $A_t \leq E_t^{(S^\star)}$ pointwise, the event $\{\exists t : A_t \geq \lambda\}$ is contained in $\{\exists t : E_t^{(S^\star)} \geq \lambda\}$, which has probability at most $1/\lambda$. □

That the minimum of e-values is an e-value for a union null is standard (Vovk & Wang, 2021); the proof above needs only the inclusion of events, so the processes $E^{(S)}$ may live in different filtrations. Faithfulness enters only through power: adjacency-faithfulness makes every $E^{(S)}$ grow for a genuinely adjacent pair, and an instance where it fails leaves the pair undecided instead of producing a wrong edge.

3.3.1 The per-set primitive

For S with $|S| = q - 1$ write the regression $X_i = \alpha^\top (1, X_S) + \beta X_j + \varepsilon$ and let S_{ii}, S_{ij}, S_{jj} be the cross-products of the residuals of X_i and X_j after regression on $(1, X_S)$, computed from the running Gram matrix of the centred data, with $p = |S|+1$. With a right-Haar prior on the nuisance parameters and $\beta \mid \sigma \sim N(0, \sigma^2 g)$ for a fixed $g > 0$, the Bayes factor is

$$E_n = (1+G)^{-1/2}\big(1 - \kappa\, \hat{r}_n^2\big)^{-(n-p)/2}, \qquad G = g\, S_{jj}, \quad \kappa = \tfrac{G}{1+G}, \quad \hat{r}_n^2 = \tfrac{S_{ij}^2}{S_{ii} S_{jj}}. \tag{3}$$

It depends on the data only through Schur complements of the Gram matrix, so a single $O(d^2)$ statistic serves every hypothesis (i, j, S). The prior scale g must be fixed before the data: a g that adapts to the running sample size destroys the martingale property. We fix it in advance and use a warm-up prefix, discarded from every process, only to standardise the variables.

Proposition 4 *Suppose $\varepsilon \sim N(0, \sigma^2)$ is independent of (X_j, X_S) and $\beta = 0$. Then (E_n) is a non-negative martingale with mean one in the filtration generated by the sequence of maximal invariants of the nuisance group (translations in the span of $(1, X_S)$ and rescaling of X_i), and therefore $\mathbb{P}(\exists n : E_n \geq \lambda) \leq 1/\lambda$ for every $\lambda \geq 1$.*

The proof is the right-Haar argument of Hendriksen, de Heide, and Grünwald (2021) and Grünwald et al. (2024) and is given in Appendix A. The regressors may have any distribution; Gaussianity is required of the regression error ε, which for $S \supseteq \mathrm{pa}(i)$ with no descendant of i is the structural error e_i.

Remark 1 (The raw-observation filtration) (E_n) is *not* a martingale in the filtration generated by the raw observations. With $S = \emptyset$, $g = 0.25$ and $X_{1:2} = Y_{1:2} = (-1, 1)$, Eq. (3) gives $E_2 = 1$ while $\mathbb{E}[E_3 \mid \mathcal{F}_2] = 1.083 > 1$ under an i.i.d. standard Gaussian null (numerical integration). Amenability of the nuisance group does not repair this. The guarantee used here, a bound on the probability of ever crossing, does not need the raw filtration, which is why Proposition 3 was stated for events.

Remark 2 (Role averaging) Eq. (3) treats X_i as the response, and the roles of i and j are interchangeable under the null. The implementation averages the two role-swapped Bayes factors to remove the arbitrary choice. A crossing of the average implies a crossing of one of the two, each of probability at most $1/\lambda$, so the guaranteed bound for the averaged process is $2/\lambda$.

Remark 3 (Beyond Gaussian errors) When the regression error is not Gaussian, or its variance depends on the regressors, Eq. (3) is the $d = 1$ case of the model-based test of Lindon et al. (2026), whose guarantee under the moment null $E[W\varepsilon] = 0$ is asymptotic and holds when the model variance equals the heteroscedasticity-robust variance. A robust sandwich version exists (Lindon et al., 2026, Theorem 5.4 of the arXiv version). Finite-sample validity over all finite-variance laws is impossible without tail control (Shah & Peters, 2020). We therefore state the adjacency guarantee as *exact for Gaussian regression errors and asymptotic otherwise*, and Section 4.3 measures the finite-sample departure under heavy tails and heteroscedasticity for both versions.

Remark 4 (One model for both certificates) Both certificates live in the linear structural equation model with independent errors. Adjacency uses only second moments, through Lemma 1; orientation uses the shape of the error laws and, in the additive-noise extension, nonlinear mechanisms, for identifiability only. The validity statement of the orientation certificate (Section 3.4) does not depend on either error law.

3.4 The orientation certificate

Fix a pair (x, y). For each direction we fit a forecaster of the joint law on the past and score it on the next observation. Let $p_s^{x\to y}$ be the density of the forecaster for the factorisation $p(x)\,p(y \mid x)$ fitted on observations $1, \dots, s-1$ and $p_s^{y\to x}$ the mirror forecaster, and set

$$d_s = \log p_s^{x\to y}(x_s, y_s) - \log p_s^{y\to x}(x_s, y_s). \tag{4}$$

Both forecasters are $\mathcal{F}_{s-1}$-measurable, and both are densities on the same space, so d_s is a legitimate score difference whether or not either model is correct. Let $\bar{\mu}_t = t^{-1}\sum_{s\le t}\mathbb{E}[d_s \mid \mathcal{F}_{s-1}]$ and let $\hat{\mu}_t$ and $\hat{\sigma}_t$ be the running mean and standard deviation of (d_s). The confidence sequence of Waudby-Smith et al. (2024, Theorem 2.2) is

$$\hat{\mu}_t \pm \hat{\sigma}_t\sqrt{\frac{2(t\rho^2+1)}{t^2\rho^2}\log\frac{\sqrt{t\rho^2+1}}{\alpha}}, \tag{5}$$

with ρ tuned to a reference time $t^* = 200$. We certify $x \to y$ at the first t at which the lower end exceeds zero and $y \to x$ when the upper end falls below zero; both are decided only after 60 scored observations.

Proposition 5 *Let $\mathcal{F}_{s-1}$ be the past, let the forecasters be $\mathcal{F}_{s-1}$-measurable, and suppose the martingale differences $d_s - \mathbb{E}[d_s \mid \mathcal{F}_{s-1}]$ satisfy the moment conditions of Waudby-Smith et al. (2024, Theorem 2.2). Then (5) is an asymptotic confidence sequence for $\bar{\mu}_t$ at level $1-\alpha$. In particular, if $y \to x$ predicts at least as well as $x \to y$ on average ($\bar{\mu}_t \le 0$ for all t), the probability of ever certifying $x \to y$ is at most α asymptotically.*

What is certified is therefore that one factorisation has a strictly better expected log score, and this is valid without either forecaster being correct. It corresponds to the

causal direction under the identifiability results for additive-noise models (Hoyer et al., 2009; Peters & Bühlmann, 2014) and for linear non-Gaussian models (Shimizu et al., 2006): if the data come from $x \to y$, the true class is contained in the forecaster family and the estimates are consistent, the expected score of the correct factorisation exceeds that of the reversed one by a positive Kullback–Leibler gap, because for generic mechanisms or non-Gaussian errors the reversed factorisation cannot represent the joint law. Under confounding or selection the better predictor need not be the cause, and the certificate then says nothing about causation. The conclusion of Proposition 5 is asymptotic, and Section 4.5 reports its finite-sample behaviour.

Forecasters.

Both use a generalised-Gaussian density $g(u) \propto \exp(-|u/s|^\kappa)$ with shape $\kappa \in [0.9, 8]$ fitted by maximum likelihood, a generalised-Gaussian marginal for the cause, and a cubic truncated-power spline mean with four quantile knots, set from the warm-up prefix, for the effect. The class `gg` has constant scale; the class `ggh` lets the log-scale be quadratic in the cause, which covers heteroscedastic noise (Immer et al., 2023; Xu et al., 2022). The *agreement* certificate reports a direction only when both classes certify the same one. Each class is Ville-bounded by the same argument, so agreement cannot err more often than either alone; its purpose is protection against model-class mistakes: the constant-scale class alone certifies the wrong direction on every heteroscedastic Gaussian pair it decides in our simulations, and the location-scale class alone is too flexible to resolve linear non-Gaussian pairs (Section 4.5). Preprocessing is fixed in advance: tied values are dequantised, the order is randomly permuted, and $\max(100, \lceil 0.2n \rceil)$ observations form a warm-up prefix used for standardisation and knots.

3.5 The algorithm and its guarantee

CERT-CD maintains a three-valued state for every pair. It streams data in batches, updates the running Gram matrix, recomputes $A_t(i, j)$ for every undecided pair, and certifies adjacency when $A_t(i, j)$ crosses its threshold. The total budget is split as $\alpha = \alpha_A + \alpha_D$, with α_A spread over the $\binom{d}{2}$ adjacency nulls and α_D over the direction nulls, and the thresholds are $\lambda_A = \binom{d}{2}/\alpha_A$ and $\lambda_D = d(d-1)/\alpha_D$, fixed before the data arrive. Which pairs are examined does not enter the bound.

Theorem 6 (Adjacency) *Under Assumption 1, linear structural equations with independent errors, and Gaussian regression errors in (3) for every examined (i, j, S), the probability that CERT-CD ever certifies an edge between a non-adjacent pair is at most $2\alpha_A$, uniformly over time and at every data-dependent stopping time of the run.*

Proof By Remark 2 each pair's averaged minimum has crossing probability at most $2/\lambda_A$ by Proposition 3. A union bound over at most $\binom{d}{2}$ non-adjacent pairs gives $\binom{d}{2} \cdot 2/\lambda_A = 2\alpha_A$. □

Choosing $\lambda_A = 2\binom{d}{2}/\alpha_A$ gives α_A exactly; the experiments use the shipped threshold, so the guaranteed bound there is $2\alpha_A$. Under non-Gaussian errors the bound

Table 1 Probability that a false edge is ever reported over the grid $n = 100, \ldots, 6400$ in 150 random eight-node graphs (Wilson 95% interval for CERT-CD, and for the others where shown in the results file).

Method	Laplace errors	Gaussian errors
CERT-CD	0.027 $[0.010, 0.067]$	0.040 $[0.018, 0.085]$
PC, $\alpha = 0.05$	0.747	0.773
PC, $\alpha = 0.01$	0.267	0.320
GES-BIC	0.913	0.960
DirectLiNGAM	0.760	1.000

holds asymptotically (Remark 3). The orientation layer inside the multivariate algorithm uses a linear generalised-Gaussian universal-inference certificate whose validity is conditional on the model describing each pair given a block of conditioning variables frozen when the edge is certified (Appendix B); the relative-fit certificate of Section 3.4 is bivariate and is evaluated on pairs. An extension that conditions the relative-fit forecasters on the frozen block is possible but not evaluated here.

4 Experiments

4.1 Protocol

The benchmarks were chosen to match the scope of the method: continuous data, linear or additive-noise mechanisms, a few hundred observations or more. The monitoring experiment (`exp13`) and the Tübingen evaluation (`exp14`) were committed to the repository with their protocols before they were run; the simulations of the pair orientation certificate (`exp16`) and of the primitive off the Gaussian model (`exp15`) were likewise written before running. Data outside the scope are reported in Section 4.7. Every number below comes from a JSON file in `results/`.

4.2 Monitoring: how often is a false edge ever reported?

We drew 150 random DAGs on $d = 8$ nodes (edge probability 0.25, coefficients with absolute value in $[0.5, 1.2]$) with Laplace and with Gaussian errors, and ran each method on nested prefixes of $n = 100, 200, 400, \ldots, 6400$ observations. A graph counts as a failure if any prefix's estimate contains an edge that is absent from the true skeleton, which is what an analyst who keeps every edge ever seen would report. CERT-CD ran as one stream with $\alpha = 0.1$ (so $\alpha_A = 0.05$ and a guaranteed bound of 0.1 in the Gaussian model), $k = 2$, warm-up of 50 rows and batches of 100. The comparators were PC (Fisher-z, $\alpha = 0.05$ and 0.01, `causal-learn` defaults), GES with the BIC score, and DirectLiNGAM with edges thresholded at $|B| > 0.05$, each rerun at every n.

CERT-CD reported a false edge in 4 of 150 graphs under Laplace errors and in 6 of 150 under Gaussian errors, against between 27% and 100% for the comparators (Table 1). The comparison is deliberately about monitoring: at a single sample size

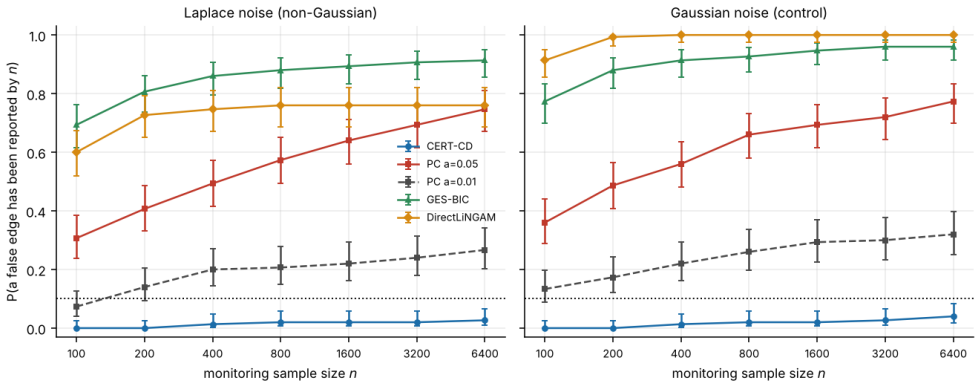


Fig. 1 Cumulative fraction of graphs with a false edge, by sample size, for each method (Laplace and Gaussian errors).

the comparators' false-edge rates are much lower, for example 42 of 150 graphs for PC at $\alpha = 0.05$ and $n = 6400$ under Laplace errors, so the gap is largely the cost of re-looking, which CERT-CD's guarantee absorbs. The price is coverage. At $n = 6400$ under Laplace errors CERT-CD's precision is 0.994 and its recall 0.938 ($F_1 = 0.965$), and it leaves 76.8% of the pairs undecided because it never certifies absence (Gaussian: 76.9%). By comparison, PC at $\alpha = 0.01$ has $F_1 = 0.970$ at the final sample size. CERT-CD's orientation is switched on by default here, which splits the budget; edges, not arrowheads, are scored.

4.3 The adjacency primitive: cost of anytime validity and calibration off-model

Cost and benefit of the sequential test.

With 2000 replications under the independence null and monitoring up to $n = 1000$, a Fisher-z test inspected at every sample size falsely rejects in 34.6% of runs at nominal $\alpha = 0.05$ (56.5% at 0.10, 81.6% at 0.20; 10.0% at 0.01). The same test at one fixed n rejects 5.6% (0.01: 1.0%), and the e-process 2.4% (0.01: 0.7%). The e-process is conservative by about a factor of two in this setting, which is the price of the time-uniform guarantee.

Off the Gaussian model.

The primitive is exact only for Gaussian regression errors. Table 2 reports the realised probability of ever crossing $1/\alpha$ over $n = 50, \ldots, 1000$ (monitored at every observation), with 1000 replications per cell, for a null pair with no covariate effect (S0), a chain $i \leftarrow k \leftarrow j$ tested given k, and a heteroscedastic null, under four error laws. A robust t-statistic confidence sequence (Lindon et al., 2026) is shown for comparison. For Gaussian and Laplace errors without heteroscedasticity the primitive stays below nominal. For skewed or heavy-tailed errors it is up to 2.5 times nominal at $\alpha = 0.01$ (0.025 for lognormal errors), and under heteroscedastic errors it exceeds nominal at

Table 2 Realised probability of ever crossing, $n = 50, \ldots, 1000$ monitored every observation (primitive, $g = 0.25$) versus a robust-t sequence checked every five observations, 1000 replications per cell.

Scenario	Errors	BF .05	BF .01	robust-t .05	robust-t .01
S0	Gaussian	0.023	0.006	0.044	0.009
S0	Laplace	0.016	0.004	0.032	0.009
S0	$t_{2.5}$	0.035	0.019	0.067	0.021
S0	lognormal	0.043	0.025	0.098	0.035
Chain	Gaussian	0.021	0.008	0.042	0.012
Chain	Laplace	0.018	0.006	0.040	0.011
Chain	$t_{2.5}$	0.026	0.013	0.076	0.035
Chain	lognormal	0.039	0.019	0.086	0.026
Het.	Gaussian	0.069	0.025	0.066	0.022
Het.	Laplace	0.065	0.028	0.053	0.017
Het.	$t_{2.5}$	0.042	0.025	0.077	0.035
Het.	lognormal	0.061	0.032	0.089	0.037

$\alpha = 0.05$ even for Gaussian errors (0.069). The robust version is not uniformly better: it is closer to nominal under heteroscedasticity but wider of the mark in the heavy-tailed cells (0.098 at $\alpha = 0.05$ for lognormal errors in S0). In the six-noise-law calibration of the earlier experiment (`exp8`, 2000 replications, $n \leq 800$) every cell is below nominal, the largest realised-to-nominal ratio being 0.95 and the median about 0.48. We therefore treat the adjacency bound as exact in the Gaussian model and as an approximation otherwise; in the stated experiments the approximation is worse under heteroscedasticity and heavy tails than the earlier calibration suggested.

4.4 Graphs that violate strong faithfulness

We generated 1780 random linear-Gaussian DAGs, of which 56.2% satisfy the separator-size premise, and stratified them by whether δ-strong faithfulness ($\delta = 0.15$) holds (1000 graphs) or fails (500 graphs). In the strata where it holds CERT-CD certified no false edge (0 of 1000) and made 8 wrong-arrow certifications (rate 0.008); where it fails it certified 6 false edges in 500 graphs (0.012, Wilson $[0.006, 0.026]$) and 29 wrong arrows (0.058, $[0.041, 0.082]$; one-sided binomial test against 0.05, $p = 0.23$). PC lost a true edge in 8 of 1000 and 241 of 500 graphs (0.482 $[0.439, 0.526]$) and our earlier delete-based algorithm, SPRINT-CD, in 0 and 169 (0.338 $[0.298, 0.381]$). These are different error types and are not commensurable: a lost edge is an omission, a false edge a commission, and CERT-CD pays for its few commissions with undecided pairs (75% undecided in this experiment; 97% of true edges certified and 94% oriented, against a CPDAG ceiling of 28% of edges orientable).

Table 3 Pair orientation, 50 simulated pairs per family, $n = 800$. Entries are (decided, wrong). Forced is the accuracy when a direction is always output.

Family	agree	gg	ggh	forced gg	forced ggh
linear, Laplace	(3,0)	(27,0)	(3,0)	0.96	0.46
linear, uniform	(0,0)	(49,0)	(7,7)	1.00	0.34
linear, Gaussian	(0,0)	(0,0)	(0,0)	0.44	0.44
ANM, tanh	(41,0)	(46,0)	(44,0)	1.00	1.00
ANM, square	(18,0)	(46,0)	(18,0)	1.00	1.00
ANM, sine	(43,0)	(50,0)	(43,0)	1.00	1.00
heteroscedastic Gaussian	(0,0)	(32,32)	(4,0)	0.06	1.00
strongly heteroscedastic	(0,0)	(9,9)	(25,0)	0.64	0.98
heteroscedastic Laplace	(39,0)	(45,0)	(40,0)	1.00	1.00

Table 4 Tübingen pairs (80 scored of 102). Accuracy is among decided pairs; interval is Clopper–Pearson 95%; p is the one-sided binomial test of accuracy $> 1/2$.

Certificate	decided	correct	accuracy	95% interval	p
agree (primary)	12	11	0.917	$[0.615, 0.998]$	0.003
gg only	23	19	0.826	$[0.612, 0.950]$	0.001
ggh only	18	14	0.778	$[0.524, 0.936]$	0.015

4.5 Orientation of pairs

Simulated pairs.

We simulated 50 pairs of $n = 800$ for each of nine families, and ran the relative-fit certificate at $\alpha = 0.05$ (Table 3). The agreement certificate committed to a wrong direction in none of the 450 pairs. The two classes alone fail in the way the design anticipated: the constant-scale class certifies the wrong direction on all 32 and 9 pairs it decides in the two heteroscedastic Gaussian families, and the location-scale class is wrong on all 7 it decides for uniform linear errors. Agreement commits on all the additive-noise families (for example 41 of 50 for the tanh mechanism) and on Laplace heteroscedastic pairs (39 of 50), declines the strongly heteroscedastic Gaussian pairs, and, as it should, declines linear Gaussian pairs, which are not identifiable.

4.6 The Tübingen cause-effect pairs

The protocol was fixed before the run: all 102 scalar-scalar pairs, official weights, tied values dequantised, a random permutation, at most 2000 rows per pair, $\alpha = 0.05$, seed 20261023. Pairs with fewer than 100 post-warm-up rows (22 of 102) cannot be scored and count as abstentions; 80 pairs are scored. This protocol was written after an earlier linear certificate decided 14 pairs and was right on 6; no parameter of the relative-fit certificate was tuned on these data.

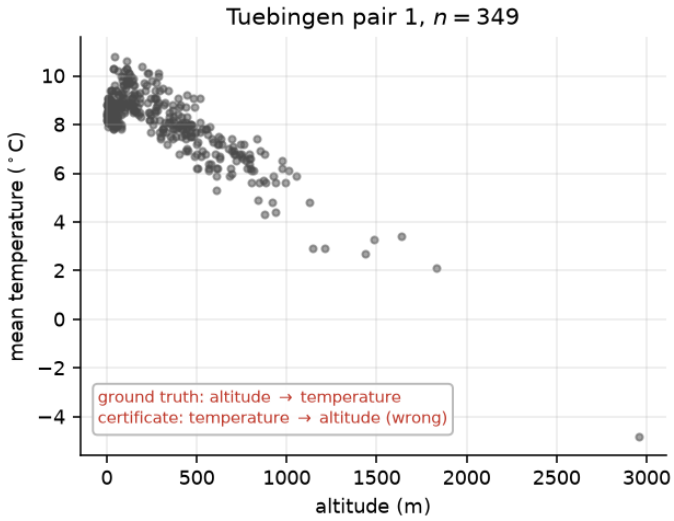


Fig. 2 Tübingen pair 1: scatter plot and the confidence sequence for the log-score difference.

The agreement certificate decided 12 of 80 scored pairs (15%, which is 11.8% of all 102) and was right on 11 (Table 4); weighted by the official weights its accuracy is 0.816. Forced to always output a direction, the constant-scale class is right on 72.5% of the 80 scored pairs (0.65 weighted) and the location-scale class on 70% (0.705). Published methods report forced accuracies of roughly 63–83% on all pairs (Immer et al., 2023; Mooij et al., 2016; Tagasovska et al., 2020), under their own protocols; abstaining methods in the literature report 75–85% at coverages of 35–85%. These numbers are not directly comparable to ours, which are on a subset, with abstention, at a pre-fixed α. The evidence is modest: 12 decisions, a wide interval, and a statement about predictive fit. Figure 2 shows pair 1, where both classes certify the direction.

4.7 Scope: where the certificates fail

Assumption violations.

We simulated 200 datasets of $n = 3000$ for each of four violations at $\alpha = 0.05$. In-model there were no false edges and no wrong arrows. Violating the separator-size premise gave a false edge in 200/200 runs and false arrows in 26; a latent common cause gave a certified edge in 200/200 runs, which is a MAG adjacency and not an error in that sense, with 38 wrong arrows; nonlinear mechanisms gave false edges in 200/200 runs and 142 wrong arrows; Poisson counts gave false edges in 200/200 runs and 28 wrong arrows. The adjacency certificate is valid only for the linear model with independent errors, and the orientation certificate inside the multivariate algorithm fails badly under nonlinearity.

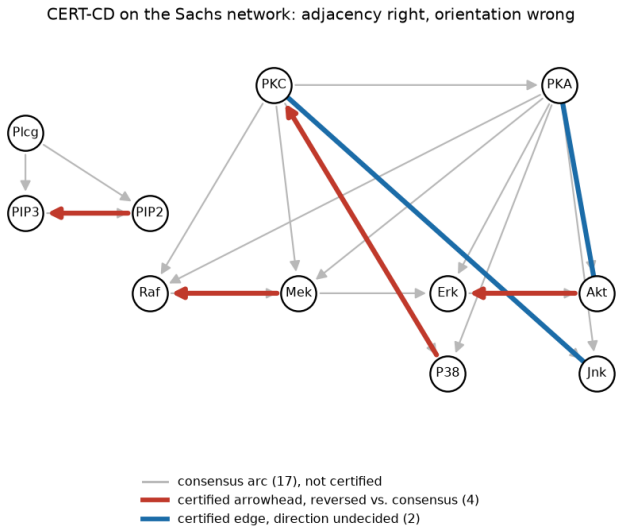


Fig. 3 Sachs data: certified edges and arrows against the consensus network.

Sachs protein-signalling data.

The data (Sachs, Perez, Pe'er, Lauffenburger, & Nolan, 2005) are discrete-like, non-linear and heavy-tailed and sit outside the method's scope, which is why we report them here. On 853 cells (753 after a 100-cell warm-up) CERT-CD certified 6 edges among 55 pairs, all in the consensus skeleton, left 49 pairs undecided, and certified 4 arrowheads, all reversed relative to the consensus network. PC found 7 edges, all in the consensus skeleton. DirectLiNGAM's causal order contradicts the consensus direction on all six pairs CERT-CD certified. Consensus directions are themselves contested (Mooij, Magliacane, & Claassen, 2020), but the certified arrows are confidently unreliable, and nothing in the output says so. A comparison with the iterative protocol of Uehara (2026) (precision 0.933, recall 0.700 with 21 queries) is not like for like: the protocol, the ground truth and the interaction model differ.

5 Discussion

What is established.

Under monitoring, the adjacency certificate kept the probability of ever reporting a false edge near 3–4% in random eight-node graphs, where the comparators reached 27% to 100%, and its stated bound is exact in the Gaussian regression model. The orientation certificate commits rarely and, on the Tübingen pairs under a frozen protocol, was right on 11 of the 12 pairs on which it committed.

Limitations.

- *The adjacency bound is asymptotic off the Gaussian model.* The null that a partial regression coefficient is zero follows from d-separation in a linear model with independent errors, but the primitive is exact only for Gaussian regression errors.

Section 4.3 measures how far from nominal it is under skewed, heavy-tailed and heteroscedastic errors; we have no finite-sample correction. Conditional independence testing without a model restriction is impossible (Shah & Peters, 2020), so none is to be expected without further assumptions.

- *Most pairs stay undecided.* About three quarters of the pairs in the eight-node experiments are undecided at $n = 6400$ because the algorithm never certifies absence, and the orientation certificate decided 15% of the scorable Tübingen pairs. Certificate-only output trades coverage for validity, and a user who needs a complete graph needs a different method.
- *Predictive fit is not causation.* The orientation certificate certifies a better expected log score. It coincides with the causal direction under additive-noise or linear non-Gaussian identifiability and under causal sufficiency; under confounding it need not, and the multivariate algorithm assumes causal sufficiency.
- *The orientation layer inside the multivariate algorithm is the old linear certificate.* Its guarantee is conditional on a linear generalised-Gaussian model and fails outside it (Section 4.7). Conditioning the relative-fit forecasters on the frozen block would remove this, and is not done here.
- *Assumed, not checked.* The linear structural model with independent errors is an assumption. Residual diagnostics on the warm-up prefix can flag gross departures, but they are not part of the guarantee.
- *Small evidence base for orientation.* The Tübingen result rests on 12 committed pairs out of 80 scorable, and 22 of the 102 pairs were too short or discrete to score; the confidence interval on the accuracy is wide. The method was designed after an earlier version failed on the same data, without tuning on it, and Section 4.6 reports the protocol that was fixed.
- *Domain.* The experiments use continuous data from mostly low-dimensional systems. The Sachs data, which are not of this kind, are reported as out-of-scope behaviour (Section 4.7).

6 Conclusion

Asserting a feature of a causal graph only by rejecting a null that is true when the feature is absent removes the dependence of the false-edge guarantee on faithfulness and makes the guarantee survive monitoring. In simulation, this holds at a few percent where deletion-based methods reach most of the graphs, with the cost that most pairs are left undecided. A predictive-fit orientation certificate extends the same discipline to direction and committed rarely but mostly correctly on a standard benchmark. Both certificates are valid only within their models, and the paper reports where they fail.

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Ethics approval. Not applicable. The study involves no human participants or animals. The protein-signalling data analysed in Section 4.7 are publicly available and were published by Sachs et al. (2005).

Data and code availability. All simulation code, experiment scripts, random seeds and figure-generating scripts are available at https://github.com/vinhqdang/SPRINT-CD. The protein-signalling data of Sachs et al. (2005) are redistributed with the `bnlearn` package (Scutari, 2010) and are downloaded by the experiment script rather than included in the repository. The Tübingen cause-effect pairs of Section 4.6 are downloaded on first use from https://webdav.tuebingen.mpg.de/cause-effect/ and are not redistributed.

Author contributions. The author is solely responsible for all aspects of this work.

Use of generative AI. A large language model was used in support of writing this manuscript. The author verified and modified the resulting content and takes full responsibility for the manuscript in its entirety, including its claims, derivations, experiments and references.

Appendix A The safe linear primitive

Setup.

Let $y \in \mathbb{R}^n$ stack the observations of the response X_i, x those of the tested covariate X_j, and $W \in \mathbb{R}^{n\times q}$ the conditioning block with an intercept, $q = |S| + 1$. The model is $y = W\alpha + \beta x + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I_n)$, with null $\beta = 0$. Let $M = I - W(W^\top W)^{-1}W^\top$ and $\tilde{y} = My$, $\tilde{x} = Mx$, and write $S_{ij} = \tilde{x}^\top\tilde{y}$, $S_{ii} = \tilde{y}^\top\tilde{y}$, $S_{jj} = \tilde{x}^\top\tilde{x}$, so that $\hat{r}^2 = S_{ij}^2/(S_{ii}S_{jj})$.

Marginal likelihoods.

The null is invariant under $y \mapsto \lambda y + Wc$, $\lambda > 0$, and the right-Haar prior is $\pi(\alpha, \sigma) \propto \sigma^{-1}$. Integrating α and σ gives $m_0(y) = C_{n,q} S_{ii}^{-(n-q)/2}$. Under the alternative, put $\beta \mid \sigma \sim N(0, g\sigma^2)$. Completing the square in β,

$$m_1(y) = C_{n,q}(1 + gS_{jj})^{-1/2}\Big(S_{ii} - \frac{S_{ij}^2}{S_{jj} + g^{-1}}\Big)^{-(n-q)/2},$$

and with $G = gS_{jj}$, $\kappa = G/(1+G)$,

$$E_n = \frac{m_1}{m_0} = (1+G)^{-1/2}\big(1 - \kappa\hat{r}_n^2\big)^{-(n-q)/2},$$

which is (3). The constant g is fixed before the monitored data arrive, from a discarded warm-up prefix.

Proof of Proposition 4.

A Bayes factor with a right-Haar prior on a group acting on the nuisance parameters of both hypotheses equals the Bayes factor of the maximal invariant (Berger, Pericchi, & Varshavsky, 1998; Hendriksen et al., 2021). The maximal invariant's law under the null does not depend on (α, σ), so E_n is a ratio of a density to its null density in the filtration $(\mathcal{G}_n)$ generated by the maximal invariants, hence a test martingale there. Ville's inequality gives $\mathbb{P}(\exists n : E_n \geq 1/\alpha) \leq \alpha$ under every Gaussian null.

The raw-observation filtration.

(E_n) is not a martingale in the filtration of the raw observations. Take $S = \emptyset$, $g = 0.25$ and $X_{1:2} = Y_{1:2} = (-1, 1)$, with the Gram matrix centred. Then $E_2 = 1$, while under an i.i.d. standard Gaussian independence null the exact conditional expectation is $\mathbb{E}[E_3 \mid \mathcal{F}_2] \approx 1.083 > 1$. The guarantee is therefore the Ville bound on the crossing event, which is what Propositions 3 and Theorem 6 use; it is not a claim about $\mathbb{E}[E_\tau]$ at arbitrary stopping times of the raw filtration.

Appendix B The linear orientation certificate inside the multivariate algorithm

Inside the streaming algorithm, direction is certified after the pair's adjacency, with a sequential universal-inference ratio (Wasserman, Ramdas, & Balakrishnan, 2020). For a pair (i, j) certified at the stopping time T, let Z_{ij} be a block of conditioning variables fixed at T. The numerator is a generalised-Gaussian linear factorisation $x \to y$ given Z_{ij} fitted sequentially on the past after T, with shape in $[0.9, 8]$ and a continuation across the Gaussian boundary of the shape grid; the denominator is the supremum over the reversed linear factorisation. The observations before T are discarded. Conditionally on $\mathcal{F}_T$ the later observations are i.i.d. and Z_{ij} is a fixed set, so the usual universal-inference argument gives an e-process for the null that the reversed linear factorisation holds; the union bound over the $d(d-1)$ ordered pairs uses the fixed family and does not depend on which hypotheses the run examines. The guarantee is conditional on the linear generalised-Gaussian model describing the pair given Z_{ij}, and it fails outside it (Section 4.7). A direction is reported after at least 80 post-certification observations.

Appendix C Inverting the primitive into a confidence sequence

Replacing y by $y - \beta_0 x$ gives the process for the null $\beta = \beta_0$, and the set of β_0 not rejected at $1/\alpha$ is the interval

$$\hat{\beta} \pm \frac{1}{S_{jj}} \sqrt{\frac{\tau(S_{ii}S_{jj} - S_{ij}^2)}{1 - \tau}}, \qquad \tau = \frac{1}{\kappa}\Big[1 - \Big(\frac{\alpha}{\sqrt{1+G}}\Big)^{2/(n-q)}\Big],$$

valid for $0 < \tau < 1$. Since $\alpha < 1$, the term in brackets is positive and $\tau > 0$ for every $n > q$; when $\tau \geq 1$ the set is the whole line. The closed form is checked in the test suite against a numerical inversion over a grid of β_0.

Appendix D Additional experimental detail

Hyperparameters.

Warm-up of 50 discarded rows for the adjacency stream; prior standard deviation 0.5 on the standardised scale ($g = 0.25$); $k = 2$; adjacency share $\alpha_A/\alpha = 0.5$; batch size

250 in the graph experiments. For the pair orientation certificate: $\alpha = 0.05$, ρ tuned to $t^* = 200$, a 60-observation minimum, warm-up $\max(100, \lceil 0.2n \rceil)$.

Reproduction.

`python experiments/run_all.py` reproduces the adjacency experiments. The scripts `exp13_monitoring_fwer.py`, `exp14_tuebingen_relfit.py`, `exp15_primitive_offmodel.py` and `exp16_pair_orientation_sim.py` reproduce the tables of Section 4; seeds are fixed in each script and the results are written to `results/` as JSON.

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